

Instantaneous, RMS and Average Values

- (i) Instantaneous value of alternating current
 $I = I_0 \sin \omega t$ or $I = I_0 \cos \omega t$
- (ii) Peak value of a.c. = I_0
- (iii) Alternating emf. = $E = E_0 \sin \omega t$ or $E = E_0 \cos \omega t$
- (iv) Mean or average value of a.c.

$$I_m \text{ or } I_a = \frac{2I_0}{\pi} = 0.637I_0 \text{ for half cycle}$$

$$= 0 \text{ for full cycle.}$$

- (v) R.m.s. value of ac $I_{\text{rms}} = I_0 / \sqrt{2} = 0.707I_0$.

Phase Difference

- (i) If the emf leads the current by $\pi/2$, the reactance is called purely inductive.
 - (ii) If the emf lags behind the current by $\pi/2$, the reactance is called purely capacitive.
- If the emf is in phase with the current, the circuit is called purely resistive.

Sign Convention

Sign for phase difference (ϕ) between I and E for series LCR circuit:

- ϕ is positive, when $X_L > X_C$
- ϕ is negative, when $X_L < X_C$
- ϕ is zero, when $X_L = X_C$

Resonance

- (i) The LCR circuit is said to be resonance when
 $X_L = X_C$ i.e., when $\omega L = \frac{1}{\omega C}$ and $\omega = \omega_0 = \frac{1}{\sqrt{LC}}$ is called resonance frequency.
- (ii) At series resonant frequency, $\omega_0 = \frac{1}{\sqrt{LC}}$, we have:
 - (a) $Z = R$ = minimum value of impedance.
 - (b) $I_0 = E_0/R$ = maximum value of peak current.
 - (c) $\phi = 0$ i.e., I and E are in phase with each other.
 - (d) V_L is equal and opposite to V_C .
 - (e) Potential drop across C and L together is zero.
 - (f) $E = V_R$

Energy Stored

- (i) Energy stored in an inductor: $U = \frac{1}{2}LI_0^2$
- (ii) Energy stored in a capacitor: $U = \frac{1}{2}CV^2 = \frac{1}{2}\frac{q_0^2}{C} = \frac{1}{2}q_0V$

Power in AC Circuit

- (i) The power in LCR circuit is given by

$$P = EI = E_0I_0 \sin \omega t \sin (\omega t - \phi).$$

Power in LCR circuit consists of two components

- (a) Virtual power component = $\frac{1}{2}E_0I_0 \cos(2\omega t + \phi)$.

It has frequency twice as that of A.C. Its value over the complete cycle is zero

- (b) Real power component = $\frac{1}{2}E_0I_0 \cos \phi$. It dissipates power and $\cos \phi$ is called power factor.

- (ii) Inductive reactance: $X_L = \omega L$

$$\text{Capacitive reactance: } X_C = \frac{1}{\omega C}$$

$$\text{Reactance: } X = X_L - X_C = \omega L - \frac{1}{\omega C}$$

Impedance

- (i) Impedance of LCR circuit: $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$$= \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

$$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{IX_L - IX_C}{IR}$$

$$= \frac{X_L - X_C}{R} = \frac{\omega L - \frac{1}{\omega C}}{R}$$

$$\text{Power, } P = E_{\text{rms}} \times i_{\text{rms}} \times \frac{R}{Z}$$

- (ii) Band width = $\omega_2 - \omega_1 = 2\Delta\omega$

$$(iii) \text{ Sharpness of resonance} = \frac{\omega_0}{2\Delta\omega} = \frac{\omega_0 L}{R}$$

$$(iv) \text{ } Q \text{ factor: } Q = \frac{\text{Voltage across } L \text{ or } C}{\text{Applied voltage}}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Transformer

- (i) $\frac{E_s}{E_p} = \frac{N_s}{N_p} = k$ (say) (transformer ratio)
- (ii) For step up transformer, $k > 1$ and for step down transformer, $k < 1$
- (iii) For step up transformer, $N_s > N_p$, therefore, $E_s > E_p$. And for the step down $N_s < N_p$ therefore, $E_s < E_p$.

- (iv) The efficiency of the transformer is given by:

$$\eta = \frac{\text{Output power}}{\text{Input power}}$$

If I_p and I_s be the currents in the primary and secondary circuit, then

$$\eta = \frac{E_s I_s}{E_p I_p}$$

For ideal transformer $\eta = 1 = 100\%$. Therefore,

$$E_s I_s = E_p I_p \text{ or } \frac{I_s}{I_p} = \frac{N_p}{N_s} = \frac{1}{k}$$

Hence, for step up transformer, current in the secondary is less than that in the primary ($I_s < I_p$) and in a step down transformer, we have $I_s > I_p$.